Student Name (in English): _____ Student ID: _____

PLO K1-CLO 2.1, 2.2, 2.3

Question 1: *MCQs* To be answered in **(30) Minutes** [ ] / **10 Marks**

Choose the best answer (if any) to complete the meaning of the following statements. **FILL THE TABLE BELOW, ONLY ONE CHOICE IS ACCEPTED.**

1	2	3	4	5	6	7	8	9	10

1. Consider the following recurrence relation:

$$T(n) = 2T\left(\frac{n}{2}\right) + n$$

With the base case $T(n) = 1$.

When solving this recurrence using the **iteration method**, what is the correct algebraic expression obtained for $T(n)$ after expanding it to the k[th] iteration step?

   A. $T(n) = 2^k T\left(\frac{n}{2^k}\right) + k.n$
   B. $T(n) = 2^k T\left(\frac{n}{2^k}\right) + \sum_{i=0}^{logn} \frac{n}{2^i}$
   C. $T(n) = 2T\left(\frac{n}{2^k}\right) + k.n$
   D. $T(n) = 2^k T\left(\frac{n}{2^k}\right) + n^k$

2. Given the following input array **A = [1, 3, 2, 5, 4, 3, 7, 6, 2, 1]**. When **Counting Sort** is applied to **A**, what will be the final content of the counter array **C** after the completion of the **third loop**?
   A. **C** = [2, 2, 2, 1, 1, 1, 1]
   B. **C** = [2, 4, 6, 7, 8, 9, 10
   C. **C** = [1, 3, 5, 6, 7, 8, 9]
   D. **C** = [2, 4, 5, 6, 8, 9, 10]

3. If a problem can be solved by combining solutions of non-overlapping sub-problems, the strategy is called:
   A. Dynamic programming
   B. Greedy
   C. Divide and conquer
   D. Brute force

4. Which of the following statements is **TRUE** regarding the **Radix Sort** algorithm?
   A. Radix Sort is a comparison-based sorting algorithm.
   B. Radix Sort starts the sorting from Most Significant Digit.
   C. Radix Sort must use a stable sorting algorithm.
   D. The space complexity of Radix Sort is always O(1).

5. Consider the recurrence: **T(n) = ½ T(n/2) + 2n** using the **Master method**, choose the correct answer.
   A. Apply case 1 and $T(n) = \Theta(n^{\log 0.5})$
   B. Apply case 2 and $T(n) = \Theta(n^{\log 0.5} \log n)$
   C. Apply case 3 and $T(n) = \Theta(n)$
   D. Master method cannot be used.

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6. Which of the following statements best describes a **Hamiltonian Cycle**?
   - **A.** A cycle that visits every edge of a graph exactly once.
   - **B.** A cycle that contains all the vertices of the graph exactly once and returns to the starting vertex.
   - **C.** A path that contains all the edges of the graph exactly once.
   - **D.** A path that connects two vertices using the minimum total weight.

7. Consider applying **Breadth-First Search (BFS)** on a graph represented using an adjacency list with V vertices and E edges. What is the time complexity of BFS?
   - **A.** $O(V^2)$
   - **B.** $O(E \log V)$
   - **C.** $O(V + E)$
   - **D.** $O(VE)$

8. Consider running **Depth-First Search (DFS)** on a graph that contains multiple disconnected components. Which of the following statements is **correct**?
   - **A.** DFS always produces exactly one DFS tree regardless of the graph structure.
   - **B.** DFS cannot be applied to disconnected graphs.
   - **C.** DFS produces multiple trees only when the graph contains cycles.
   - **D.** DFS may produce more than one DFS tree, resulting in a DFS forest.

9. At each step of **Dijkstra's Algorithm**, which vertex is selected next?
   - **A.** The unvisited vertex with the largest degree.
   - **B.** The unvisited vertex with the smallest distance from the source.
   - **C.** A random unvisited vertex.
   - **D.** The most recently visited vertex.

10. When **Dijkstra's Algorithm** terminates, what is the resulted information?
    - **A.** Only the shortest path to one destination.
    - **B.** The minimum spanning tree of the graph.
    - **C.** All cycles in the graph.
    - **D.** The shortest distance from the source to every reachable vertex.

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Question 2: *NP-completeness* To be answered in (30) Minutes [ ] / 8 Marks

Use the following words to fill in the blanks below (Each word may be used more than once). [0.5 Marks for each part].

P / NP / *NP-Complete* / *NP-Hard* / Nondeterministic polynomial / shortest / longest
Non polynomial / Heuristics / Approximation algorithms / P = NP / Graph coloring problem
Sorting / Brute force / $A \leq_p B$ / $B \leq_p A$ / Minimization / Maximization / CIRCUIT-SAT

1. _____ is an intractable problem.
2. The problem of finding the lowest production costs through better selection of existing resources is a type of _____ problem.
3. Any tractable problem is a _____ problem.
4. _____ is a tractable problem.
5. Any P problem is an _____ problem.
6. NP stands for _____.
7. It has not been proven yet that _____.
8. If $A \leq_p B$ and A is *NP*, B is _____.
9. Finding the _____ path problem is a P problem, while finding the _____ path is an NP-Complete problem.
10. If $A \leq_p B$ and A is *NP-Complete*, then B is _____.
11. To prove a problem A is a NP-complete problem, we need to find a known NP-complete problem B where _____.
12. Let A be a problem that belongs to the class of NP. If A is not NP-hard, then it is not _____.
13. If we find an algorithm to solve the 3-SAT problem in polynomial time, then Hamiltonian cycle problem will become a _____ problem.
14. The first NP-complete problem is the _____.
15. To solve NP-complete problem, use _____ that is guaranteed to be within some factor of the optimum.
16. Do not use _____ approach to solve the 3-SAT problem in reasonable time.

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PLO S1-CLO 2.3

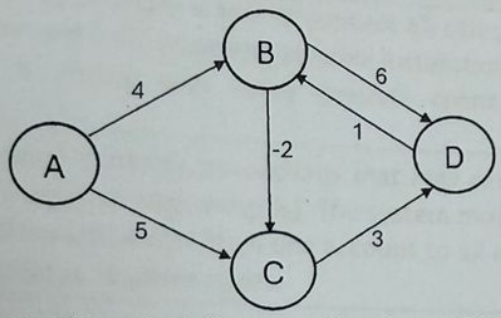
Question 3: ***Graph***

To be answered in (45) Minutes [ ] / 18 Marks

A. Consider the following directed graph: **[5 Marks]**

Apply **Bellman-ford algorithm** to find the single source shortest path from vertex **A** to all other vertices using the following **order** of the edges:

BC → BD → DB → CD → AB → AC



Show the details by filling the following table **after EACH pass (you need to draw a table for each pass).**

Vertex	A	B	C	D
D				
π				

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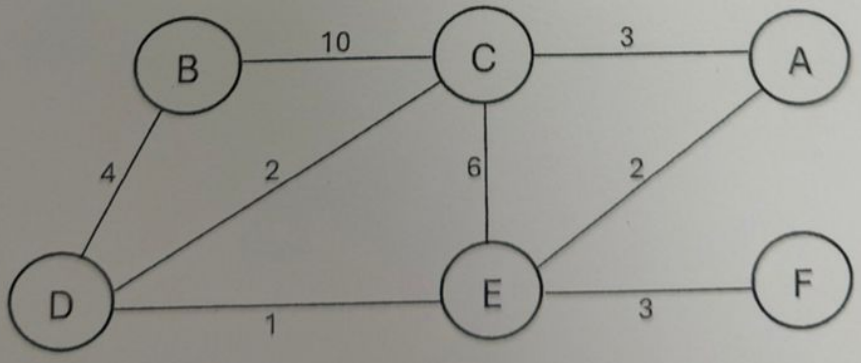
B. For each of the following scenarios, choose the **most suitable** graph algorithm from the following list:

BFS, DFS, Prim's, Kruskal's, Dijkstra's, or Bellman-Ford

[5 Marks: one mark for each part]

	Scenario	Write the Appropriate Graph Algorithm
1	An internet service provider wants to connect all cities using fiber-optic cables with the minimum possible installation cost. The network is dense, with many possible connections between cities.	
2	A financial network contains transactions that may result in gains or losses (negative edge weights). The system must find the minimum transaction cost from one account to all others and detect suspicious negative cycles.	
3	A cybersecurity analyst is investigating a computer network to determine whether a compromised machine can reach a sensitive server. The analyst explores each connection as deeply as possible before backtracking.	
4	A government is building a railway system connecting all towns with the minimum total construction cost. The list of possible railway segments is already sorted by cost.	
5	A ride-sharing application needs to find the cheapest route from a driver's current location to every other location in a city where all road costs are positive.	

C. Consider the following undirected graph.



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Build **minimum cost spanning tree** (MCST) using:

1. **Kruskal's algorithm**: Specify the **order** in which edges get added to the spanning tree, draw the resulted MCST, and determine its **cost**. **[5 Marks]**

2. **Prim's algorithm:** Determine the **order** in which vertices removed from the queue, draw the resulted MCST, and determine its **cost**. Start with the vertex **A** and use alphabetical order to break ties. **[3 Marks]**

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PLO S2-CLO 2.4

Question 4: ***Linear Programming*** **To be answered in (15) Minutes** [ ] / 4 Marks

Given the following **linear programming** problem:

Maximize $z = 5x_1 + 3x_2$

Subject to:

$2x_1 + 4x_2 \leq 8$

$x_1, x_2 \geq 0$

A. Apply the **Simplex algorithm** to find the values of x_1 and x_2 that optimize the objective value z. Show all the details.

B. What are the values of x_1, x_2, and z.